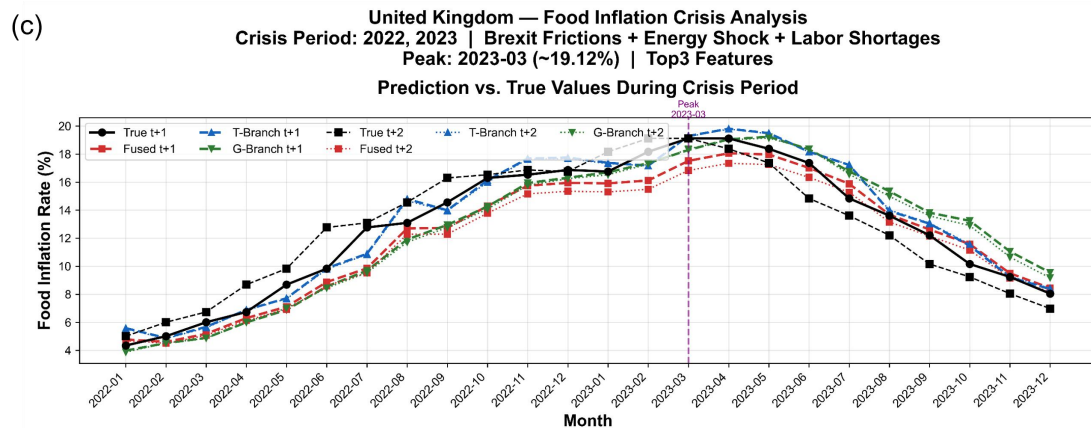
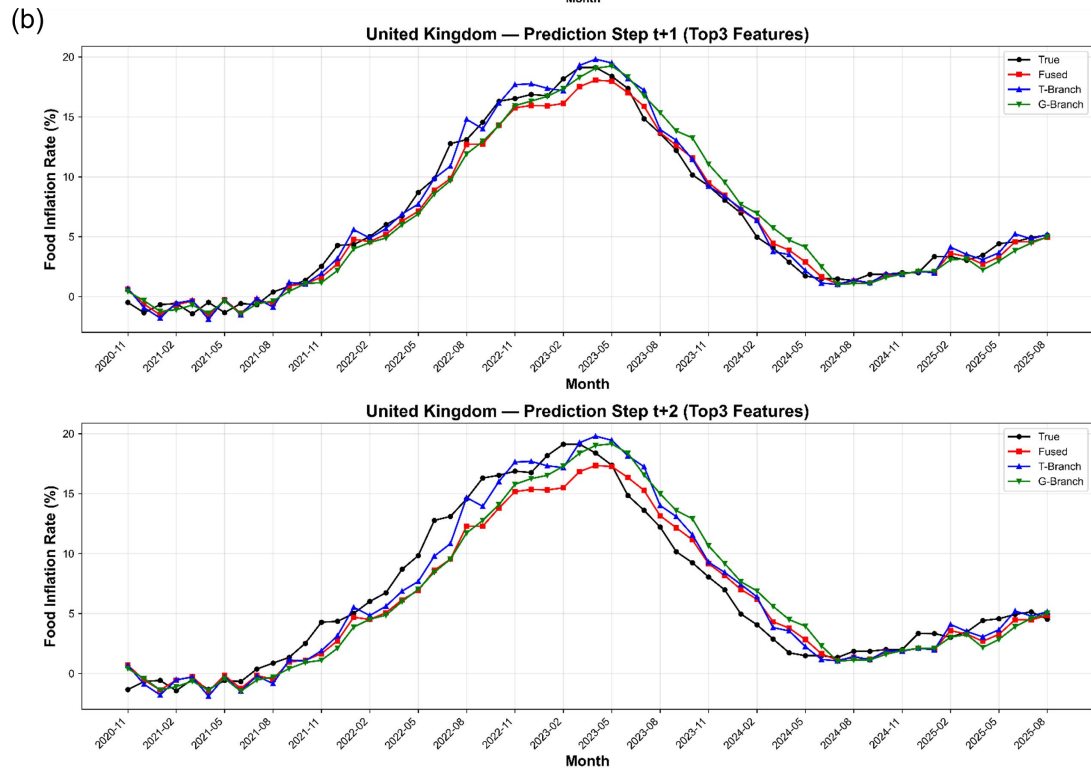
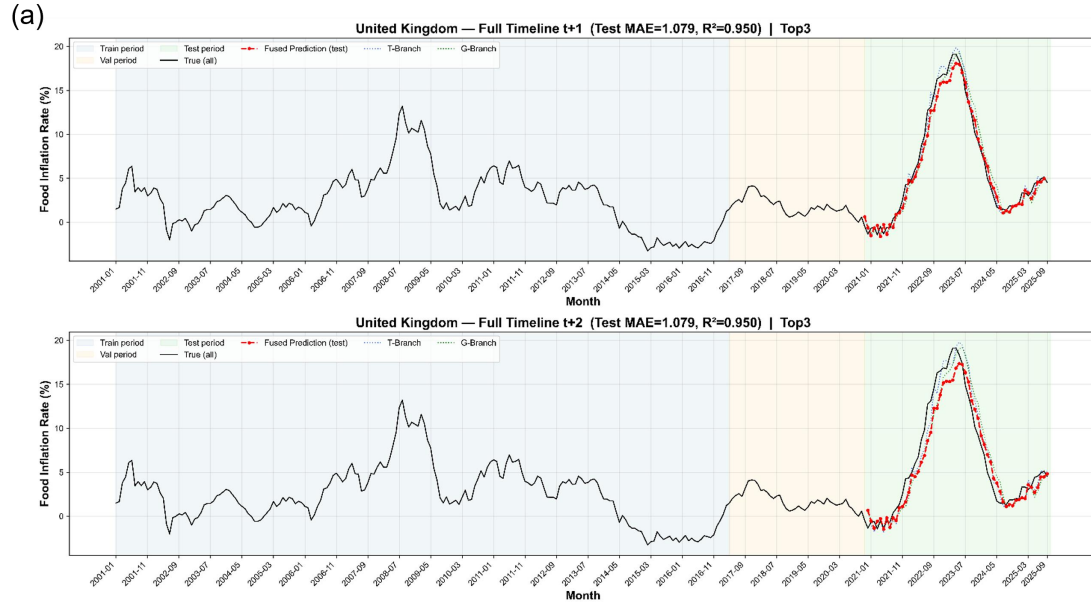
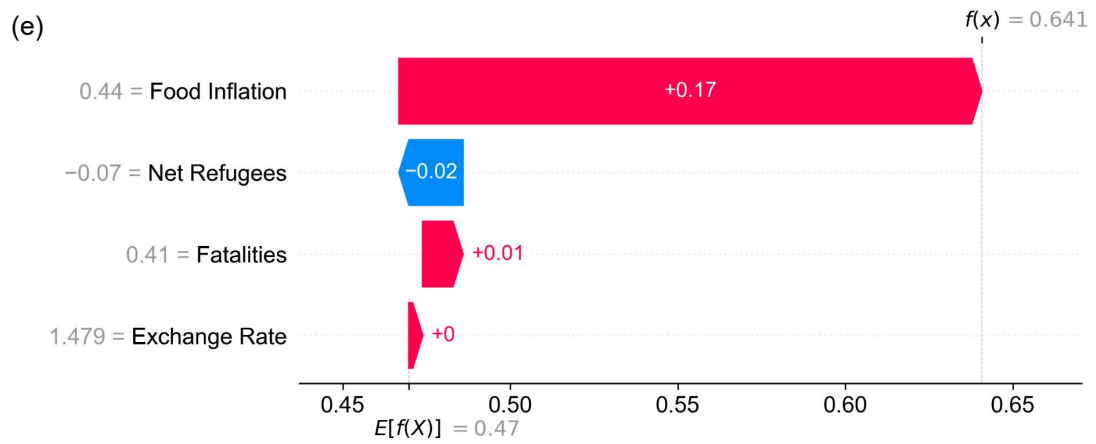
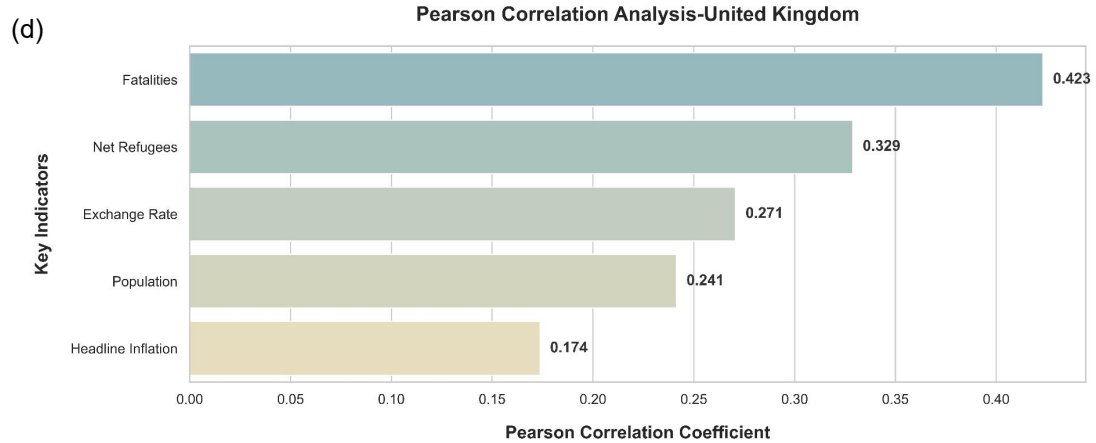






Multidimensional Evaluation and Attribution Analysis of Food Inflation Forecast in United Kingdom. (a) Overview of the total cycle prediction and fitting, (b) Multi-step Short-Term Forecast Comparison Chart, (c) Details of Prediction Performance in Extreme Crisis Periods, (d) Pearson correlation analysis of macroeconomic drivers, (e) SHAP-based explanation of feature contributions for a specific prediction instance.

Operating with a noticeable separation from traditional macroeconomic aggregates, food inflation in the United Kingdom exhibits a localized, volatility-driven pricing structure. Long-term historic tracking points to relatively constrained linear correlations between food price movements and domestic indicators, with core variables encompassing headline inflation and population shifts demonstrating limited long-run statistical alignment. During specific high-risk observation windows, structural decomposition registers a prominent positive marginal contribution from standalone food price series, suggesting that localized agricultural market disruptions potentially act as immediate risk accelerators rather than as trailing indicators. This specific configuration reflects a highly responsive import-dependent transmission profile, further conditioned by external global trade friction variables that structurally separate domestic food retail dynamics from the broader national economic performance.